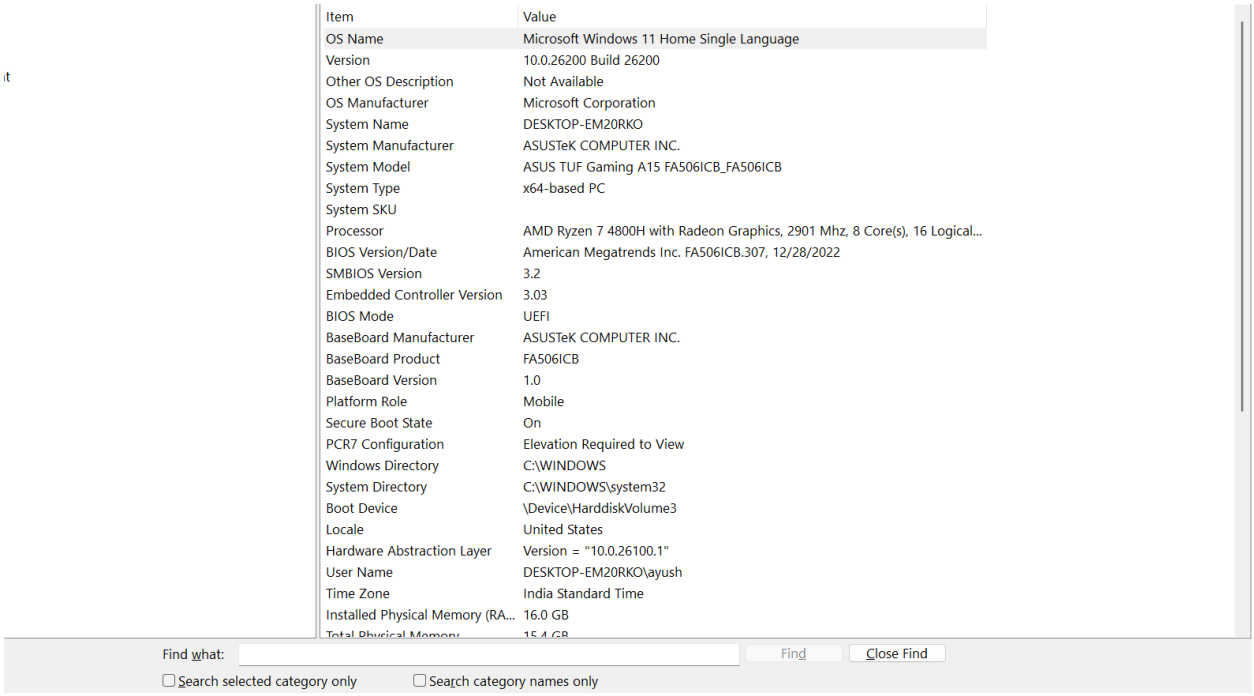
1. **On your Windows laptop or PC, open 'System Information' and take a screenshot showing your OS version, processor, and RAM details**



Item	Value
OS Name	Microsoft Windows 11 Home Single Language
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-EM20RKO
System Manufacturer	ASUSTeK COMPUTER INC.
System Model	ASUS TUF Gaming A15 FA506ICB_FA506ICB
System Type	x64-based PC
System SKU	
Processor	AMD Ryzen 7 4800H with Radeon Graphics, 2901 Mhz, 8 Core(s), 16 Logical...
BIOS Version/Date	American Megatrends Inc. FA506ICB.307, 12/28/2022
SMBIOS Version	3.2
Embedded Controller Version	3.03
BIOS Mode	UEFI
BaseBoard Manufacturer	ASUSTeK COMPUTER INC.
BaseBoard Product	FA506ICB
BaseBoard Version	1.0
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume3
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	DESKTOP-EM20RKO\ayush
Time Zone	India Standard Time
Installed Physical Memory (RA...	16.0 GB
Total Physical Memory	15.4 GB

Find what: Find Close Find

Search selected category only Search category names only

2. **Download any official Linux distribution ISO (like Ubuntu or Fedora), and use Rufus or BalenaEtcher to create a bootable USB drive for installation.

Hint: If you don't have a spare USB, use a virtual machine tool like VirtualBox to mount the ISO and start the installation process virtually.**

No.	Item	Purpose
1	Windows laptop/PC	Used to create the bootable USB drive.
2	USB drive (8 GB or larger)	Stores the Linux installation files.
3	Ubuntu ISO file	Contains the Linux operating system.
4	Rufus software	Converts the USB drive into a bootable USB.
5	Internet connection	Required to download Ubuntu and Rufus.

Steps:

1. Open the official Ubuntu website.
2. Select Ubuntu Desktop LTS.
3. Download the ISO file.
4. Save it in your Downloads folder.

Step	Action	Description
1.	Insert USB	Connect an empty USB drive of at least 8 GB to the computer.
2.	Open Rufus	Run the Rufus application.
3.	Select Device	Choose your USB drive from the Device dropdown.
4.	Select ISO	Click SELECT and choose the downloaded Ubuntu ISO file.
5.	Partition Scheme	Choose GPT for most modern UEFI computers.
6.	Start	Click START to begin creating the bootable USB.
7.	Confirm	Accept the warning that all data on the USB will be erased.
8.	Finish	Wait until Rufus displays READY, then close the application.

5. Test the bootable USB (optional)

1. Keep the USB connected and restart your computer.
2. Open the boot menu using the appropriate key (often F12, Esc, or F9).
3. Select the USB drive.
4. Wait for the Ubuntu startup screen.
5. Select Try Ubuntu to test Linux without installing it on your laptop.

6. Expected result

- Ubuntu ISO downloads successfully.
- Rufus recognizes the USB drive.
- The bootable USB is created successfully.
- Ubuntu starts when booting from the USB drive.

3. **On a macOS device (or using a macOS virtual machine), open 'System Preferences' and change the desktop wallpaper, then adjust the system language to Hindi or Gujarati.

Hint: Find 'Language & Region' in System Preferences to add or change languages.**

1. Change the desktop wallpaper

Step	Action	Description
1.	Open System Settings	Click the Apple menu in the top-left corner and select System Settings.
2.	Open Wallpaper	Find and click the Wallpaper option.
3.	Choose wallpaper	Select any available desktop background or click Add Photo to choose your own image.
4.	Apply wallpaper	Select the image and check whether it appears as your desktop background.

2. Change the system language to Hindi or Gujarati

Step	Action	Description
1.	Open General Settings	Go to System Settings → General.
2.	Select Language & Region	Click Language & Region to open language settings.
3.	Add a language	Click the (+) button under Preferred Languages.
4.	Select Hindi	Search for Hindi and click Add.
5.	Set primary language	Move Hindi to the top of the Preferred Languages list.
6.	Restart Mac	Restart the computer if macOS asks you to apply the language change.

Note: Hindi is available as a macOS system language. Gujarati may be available for typing but might not be supported for the complete system interface.

Apple Support+1

3. Expected result

- Desktop wallpaper changes successfully.
- Hindi is added to the preferred languages list.
- macOS displays supported interface elements in Hindi after applying the change.

4. **Install a fresh copy of Ubuntu Linux (real or virtual machine) and create a new user account named 'gamer'. After installation, enable automatic login for this user and take a screenshot of the login screen showing the username.**

1. Requirements

No.	Requirement	Purpose
1	Ubuntu ISO file	Contains the Ubuntu operating system.
2	VirtualBox	Allows Ubuntu to run inside Windows without replacing Windows.
3	Minimum 4 GB RAM recommended for the VM	Helps Ubuntu run smoothly.
4	Virtual disk (25 GB or more recommended)	Provides storage for Ubuntu and its files.
5	Internet connection	Helps download Ubuntu and updates.

2. Install Ubuntu Linux

Step	Action	Description
1.	Download Ubuntu	Download the Ubuntu Desktop ISO from the official website.
2.	Open VirtualBox	Create a new virtual machine and select Linux as the operating system.
3.	Allocate resources	Assign around 4 GB RAM and create a virtual hard disk of at least 25 GB.
4.	Attach ISO	Select the downloaded Ubuntu ISO as the virtual machine's installation disc.

5.	Start installation	Start the virtual machine and select Install Ubuntu.
6.	Complete installation	Follow the installation instructions and restart the virtual machine when prompted.

3. Create a new user account named gamer

During installation, Ubuntu may ask you to create an account.

Enter the following details:

Field	**Information**
Your name	Gamer
Computer name	gamer-pc
Username	gamer
Password	Create a strong password
Confirm password	Enter the same password again

Click Continue to complete the installation.

Alternatively, after installation, open Settings → System → Users, unlock the settings, and select Add User. Enter gamer as the username and create a password.

4. Enable automatic login

Follow these steps:

1. Open Ubuntu Settings.
2. Select System → Users (on some versions, Users appears directly in Settings).
3. Click Unlock and enter your administrator password.
4. Select the gamer account.
5. Find the Automatic Login option.
6. Turn Automatic Login ON.
7. Close Settings.

Now Ubuntu should automatically log in to the gamer account whenever the computer starts.

5. Verify the account

Open Terminal and enter:

whoami

Expected output:

gamer

This confirms that you are logged in using the gamer account.

6. Screenshots for practical submission

Capture these screenshots as evidence:

No.	Screenshot	What it should show
1	Ubuntu installation	Ubuntu installation screen.
2	User creation	Username field containing gamer.
3	Login screen	gamer account displayed before enabling automatic login.
4	Automatic Login setting	Automatic Login switched ON for gamer.
5	Ubuntu desktop	Desktop after automatic login.
6	Terminal	Output of the whoami command showing gamer.

Important

Once automatic login is enabled, Ubuntu bypasses the login screen. Therefore, take the login-screen screenshot before enabling automatic login, and take another screenshot afterward showing the enabled setting.

7. Expected result

- Ubuntu Linux is installed successfully.
- A new user account named gamer is created.
- Automatic login is enabled for the gamer account.
- Ubuntu opens directly into the gamer desktop after restarting.

5. **Use ChatGPT to generate a step-by-step guide for installing Windows 10 on a laptop, including partitioning tips and post-installation settings for privacy and updates. Follow the guide and note any steps that were unclear or missing.**

Practical: Installing Windows 10 on a Laptop

Important – Windows 10 support ended

Microsoft ended regular Windows 10 support on October 14, 2025. Windows 10 can still be installed and used, but it no longer receives regular free security updates. Microsoft offers an Extended Security Updates (ESU) programme for eligible devices until October 12, 2027. For a daily-use laptop, Windows 11 is the safer recommendation when compatible. For this practical, the following steps explain Windows 10 installation.

Microsoft Support+1

1. Requirements

No.	Item	Purpose
1	Laptop or PC	Device on which Windows will be installed.
2	USB drive (8 GB or larger)	Stores Windows installation files.
3	Windows 10 ISO	Contains the Windows operating system.
4	Rufus or Media Creation Tool	Creates a bootable USB drive.
5	Internet connection	Downloads Windows and updates.
6	Windows product key or digital licence	Activates Windows.
7	Backup storage	Keeps personal files safe before installation.

2. Prepare the laptop

1. Back up important files, photos, documents and videos to an external drive or cloud storage.
2. Save your Windows activation details and BitLocker recovery key, if device encryption is enabled.
3. Connect the laptop to its charger.

4. Check whether the laptop supports Windows 10 and obtain any required Wi-Fi, storage or chipset drivers from the manufacturer's website.
5. Insert an empty USB drive of at least 8 GB.

3. Create a bootable Windows 10 USB

Step	Action	Description
1	Download the tool	Download Microsoft's Windows 10 Media Creation Tool.
2	Open the tool	Run the downloaded application.
3	Accept terms	Read and accept the licence terms.
4	Select option	Choose Create installation media for another PC.
5	Choose settings	Select language, Windows 10 edition and 64-bit architecture as appropriate.
6	Select USB	Choose USB flash drive and select the correct device.
7	Create media	Allow the tool to download Windows and prepare the USB drive.

Note: Creating the bootable USB erases its existing contents. Double-check that you selected the correct USB drive.

Microsoft Support

4. Start Windows 10 installation

Step	Action	Description
1	Restart laptop	Keep the bootable USB connected and restart the laptop.
2	Open Boot Menu	Press the appropriate key, such as F12, Esc, F9 or F8, depending on the laptop manufacturer.
3	Select USB	Choose the USB drive as the boot device.
4	Choose language	Select language, time format and keyboard layout.

5	Start installation	Click Install now.
6	Enter product key	Enter your valid Windows product key or select I don't have a product key if you have an existing digital licence.
7	Choose edition	Select the edition matching your licence, such as Windows 10 Home or Pro.
8	Select installation type	Choose Custom: Install Windows only for a clean installation.

5. Partitioning the hard drive

Partitioning means dividing one physical storage drive into separate sections to organise data.

Partition	**Purpose**
C: Drive	Stores Windows, installed applications and system files.
D: Drive	Stores personal files, documents, games and other data.
EFI Partition	Contains files required to start Windows on UEFI systems.
Recovery Partition	Contains recovery tools used to repair Windows.

Partitioning tips:

1. For a laptop with a 512 GB SSD, you could allocate around 180–200 GB to C: and use the remaining space for D:.
2. Avoid creating too many partitions.
3. Keep enough free space on C: for Windows updates and applications.
4. Windows Setup can automatically create the required EFI and recovery partitions.
5. Always double-check the selected disk and partitions before deleting or formatting anything.

Data loss warning

Deleting or formatting a partition permanently removes its data. For a complete clean installation, delete partitions only on the intended target disk after backing up everything. If you want to preserve an existing D: drive or recovery partition, do not delete it. Selecting unallocated space on the correct disk is generally the simplest clean-install option.

6. Complete Windows installation

1. Select the correct unallocated space or intended Windows partition.
2. Click Next to begin installation.
3. Wait while Windows copies files and installs necessary components.
4. The laptop will restart several times.
5. Remove the USB when the first restart begins if necessary, so the laptop boots from its internal drive rather than starting the installer again.
6. Complete the initial Windows setup by selecting your region, keyboard layout and account options.

7. Post-installation privacy settings

After Windows starts, open Settings → Privacy.

Setting	Recommended action	Reason
Location	Turn off if not needed.	Prevents unnecessary location access.
Camera	Allow selected apps only.	Protects camera privacy.
Microphone	Allow selected apps only.	Prevents unnecessary microphone access.
Advertising ID	Turn off where available.	Reduces personalised advertising.
Diagnostic data	Choose Required rather than Optional, where available.	Limits optional diagnostic information sharing.
Tailored experiences	Turn off.	Avoids personalised suggestions based on diagnostic data.
Background apps	Disable unnecessary apps.	Reduces unnecessary background activity.

Microsoft provides these controls under Windows 10's Privacy settings.

Microsoft Support+2

8. Windows updates and drivers

Open Settings → Update & Security → Windows Update.

Step	Action	Purpose
1	Check for updates	Check for updates available to your device.
2	Install eligible updates	Install available security and system updates.
3	Restart laptop	Complete updates that require restarting.
4	Update graphics driver	Install the correct Intel, AMD or NVIDIA driver.
5	Check Device Manager	Look for missing drivers or yellow warning icons.
6	Check activation	Open Settings → Update & Security → Activation.

Important: Because regular Windows 10 support has ended, ordinary Windows Update does not provide the same ongoing security protection as a supported Windows release. Check eligibility and enrolment in Microsoft's ESU programme if you need to continue using Windows 10.

Microsoft Support+1

9. Expected result

- Windows 10 installs successfully.
- The laptop starts from its internal storage.
- The required partitions are created.
- Privacy settings are configured.
- Available updates and hardware drivers are checked.
- Windows activation status is verified.

10. Unclear or missing steps noticed during the process

These are common points that may need additional explanation while performing the installation.

No.	Possible confusion	Explanation
1	Which boot key should I press?	It depends on the laptop brand and model.

2	GPT or MBR?	GPT is generally used with modern UEFI systems; MBR is associated with older Legacy BIOS systems.
3	Which partition should I select?	Always identify the correct target drive by its capacity before installing.
4	Why is the Wi-Fi not working after installation?	The wireless network driver may need to be installed manually.
5	Why is Windows not activated?	The installed edition may not match the digital licence, or a valid licence may be missing.
6	Why are updates not appearing?	Windows 10 has reached the end of regular support; ESU eligibility and enrolment matter.

11. Screenshots to attach to your practical report

No.	Screenshot	Evidence
1	Bootable USB creation	Windows installation USB prepared successfully.
2	Windows Setup screen	Language and keyboard selection.
3	Product key screen	Windows activation setup.
4	Partition selection screen	Available drives and partitions.
5	Windows installation progress	Operating system installation.
6	Windows desktop	Successful installation.
7	Privacy settings	Selected privacy options.
8	Windows Update	Update settings and status.
9	Activation settings	Windows activation status.